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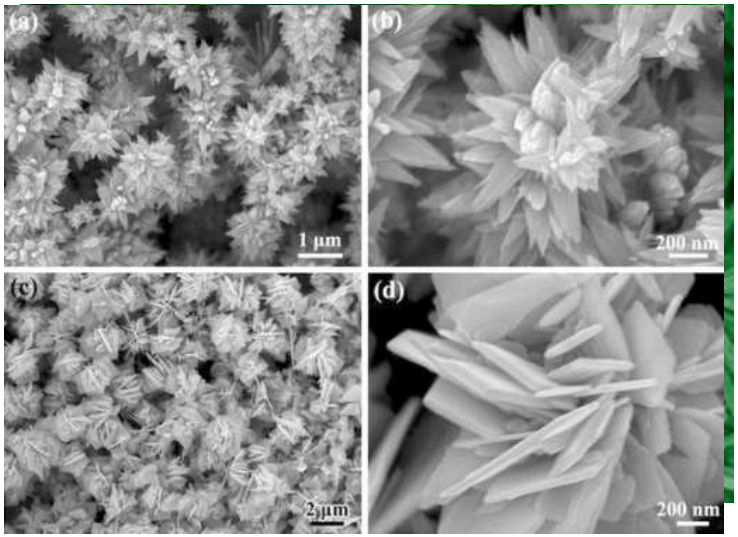
ENGINEERING CHEMISTRY (21CH12/22)
Unit-IV Chemistry of nanomaterials

Introduction to Nanomaterials/Science/Technology

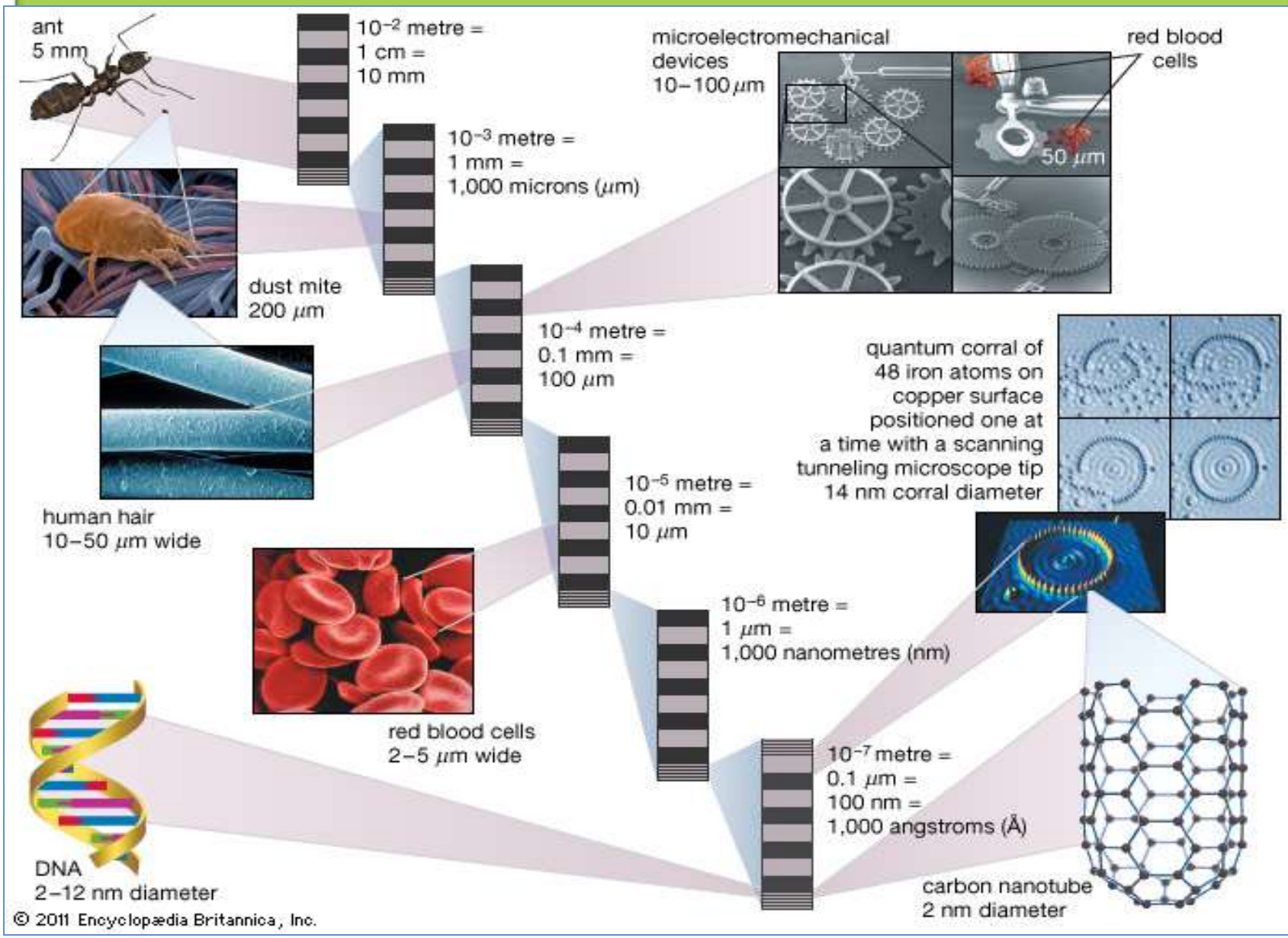


❖ **Nanomaterials** are commonly defined as materials with an average grain size less than 100 nanometers.

❖ The word ***"nano***" originates from the Greek word "nanos" which means "dwarf". However, in scientific language it is a prefix which has a value equal to "one billionth, i.e. 10^{-9}.

❖ Therefore, one nanometer is ***one billionth of a meter (1 nm = 10^{-9} m).***

Nano-dimension comparison



Nanoscience and Nanotechnology

Nanostructure science and technology is a broad and interdisciplinary area of research and development activity that has been growing explosively worldwide in the past few years. It has the potential for revolutionizing the ways in which materials and products are created and the range and nature of functionalities that can be accessed. It is already having a significant commercial impact, which will assuredly increase in the future.

"***Nanoscience*** *is the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales, where properties differ significantly from those at a larger scale"*

Nanotechnologies *are the design, characterization, production and application of structures, devices and systems by controlling shape and size at nanometre scale.*

Classification of Nanomaterials

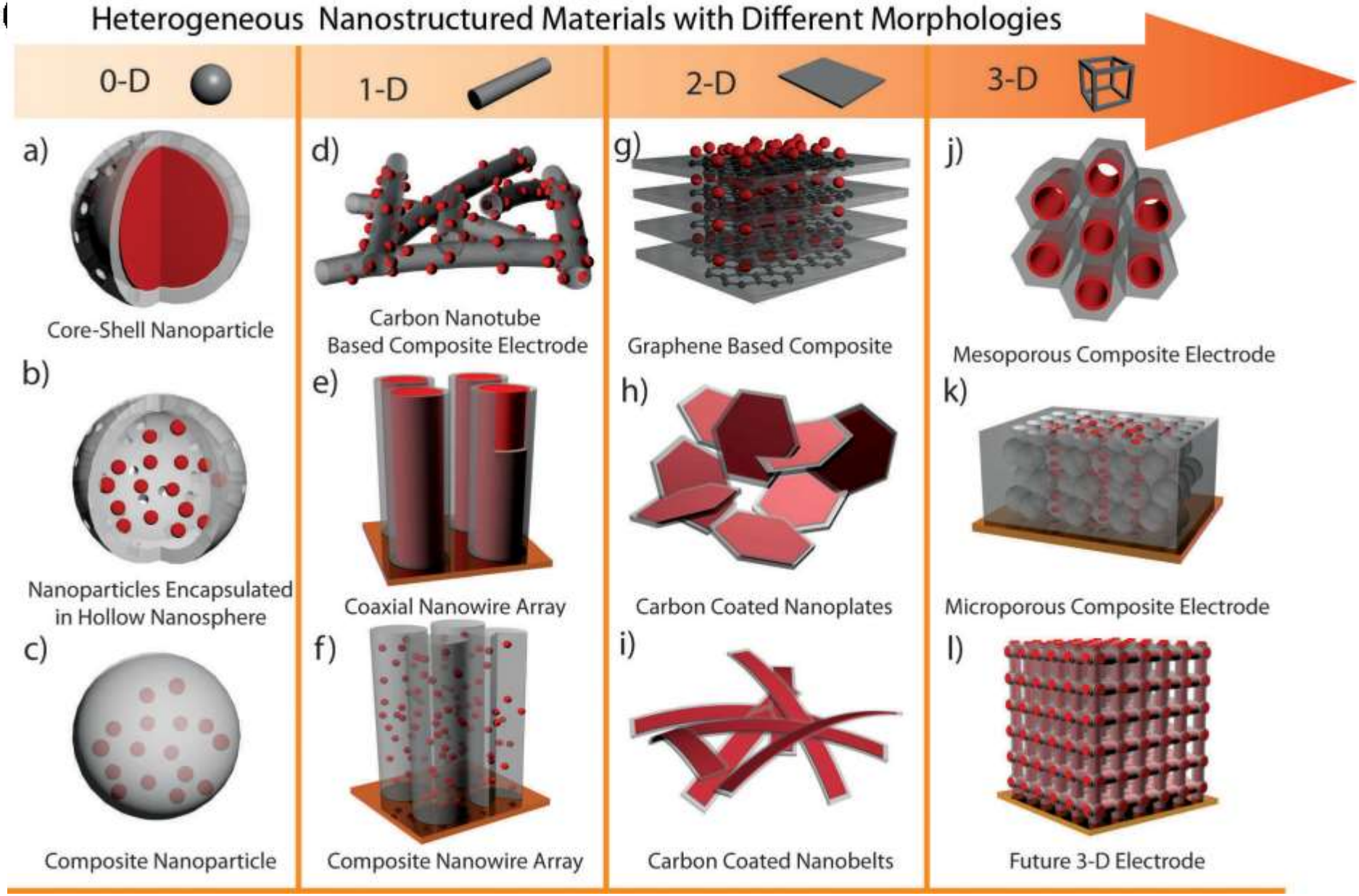
Nanostructured materials are classified as

a) ***Zero dimensional***: Nanomaterials are less than 100nm in all three dimensions.

b) One dimensional: Nanomaterials are less than 100nm in any two dimensions.

c) Two dimensional: Nanomaterials are less than 100nm in any one dimensions, and

d) Three dimensional nanostructures: Nanomaterials are aggregations of any of the above (0, 1, 2D) dimensional nanomaterial with greater than 100nm in all the three dimensions.


Heterogeneous Nanostructured Materials with Different Morphologies
0-D
1-D
2-D
3-D
a)
Core-Shell Nanoparticle
b)
Nanoparticles Encapsulated in Hollow Nanosphere
c)
Composite Nanoparticle
d)
Carbon Nanotube Based Composite Electrode
e)
Coaxial Nanowire Array
f)
Composite Nanowire Array
g)
Graphene Based Composite
h)
Carbon Coated Nanoplates
i)
Carbon Coated Nanobelts
j)
Mesoporous Composite Electrode
k)
Microporous Composite Electrode
l)
Future 3-D Electrode

Properties of Nanomaterials in comparison to bulk: Surface area, Optical, Catalytic properties

Properties of Nanomaterials

Nanomaterials have the structural features in between of those ***of atoms and the bulk materials.*** The properties of materials with nanometer dimensions are significantly different from those of atoms and bulks materials. This is mainly due to the nanometer size of the materials which render them:

(i) large fraction of surface atoms;

(ii) high surface energy;

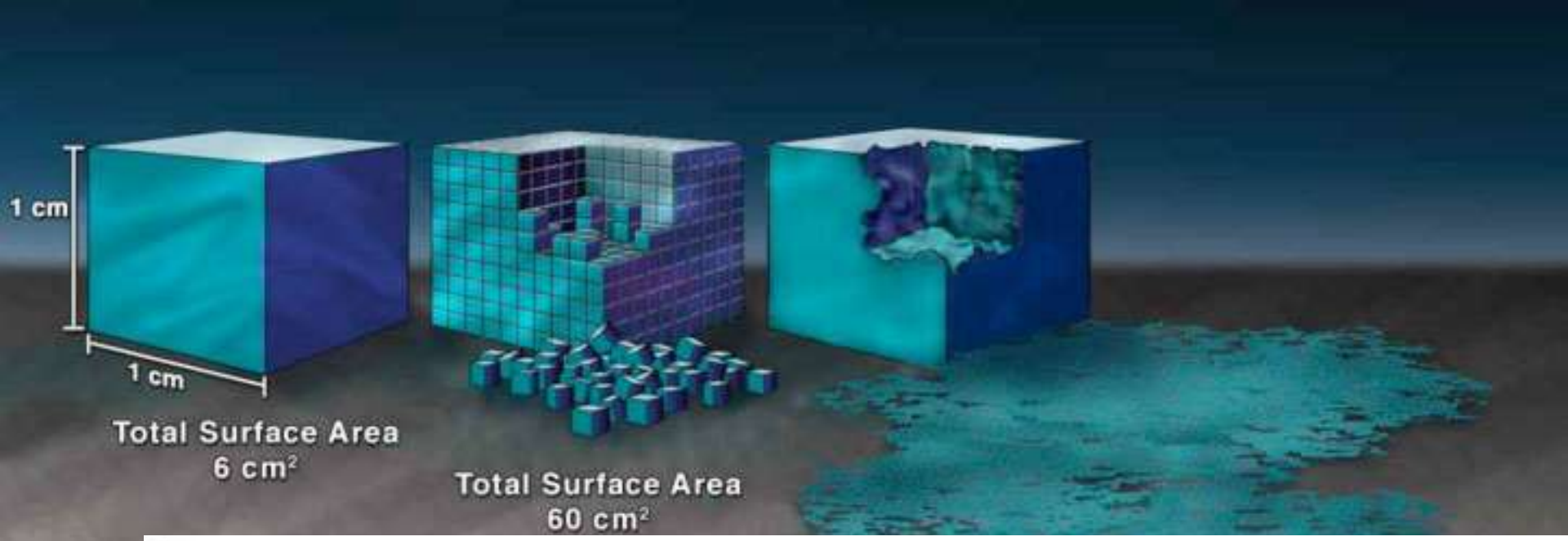
(iii) spatial confinement;

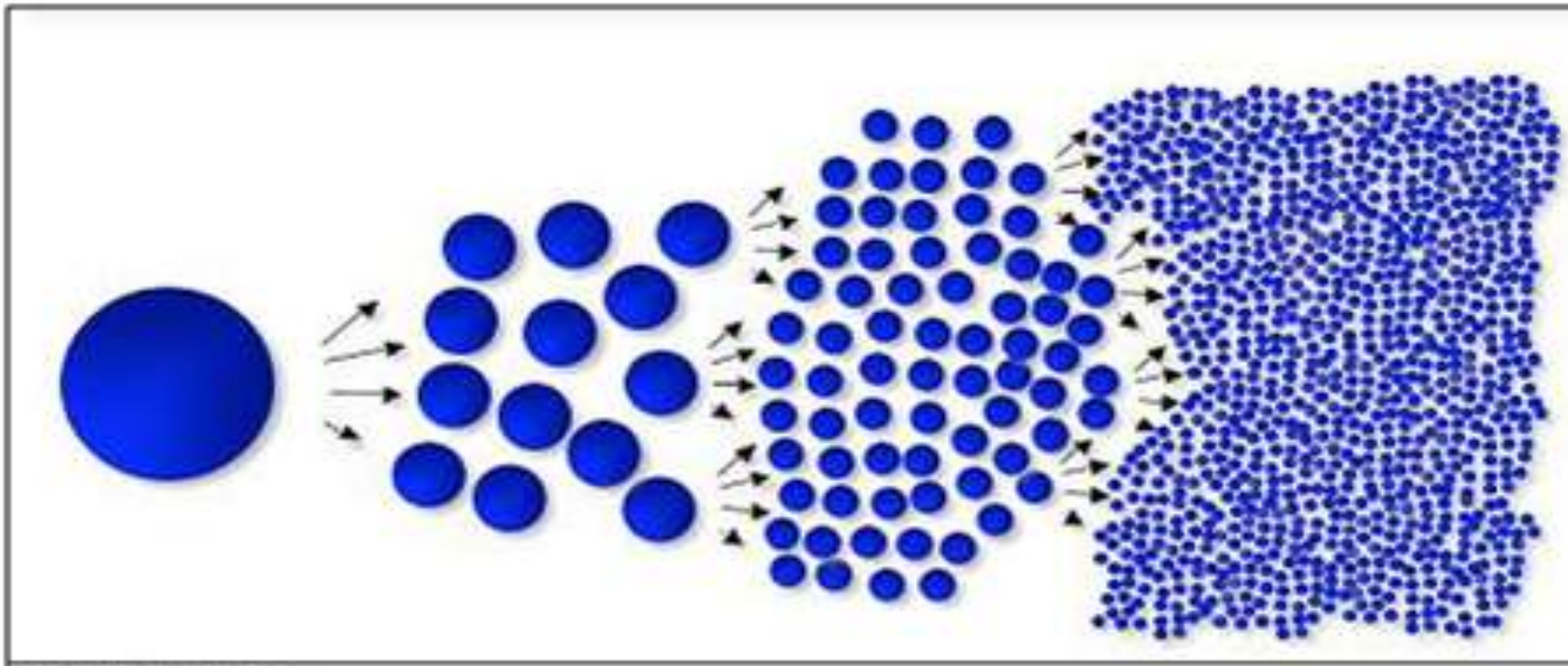
(iv) reduced imperfections,

which do not exist in the corresponding bulk materials.

Surface area of Nanomaterials and bulk materials

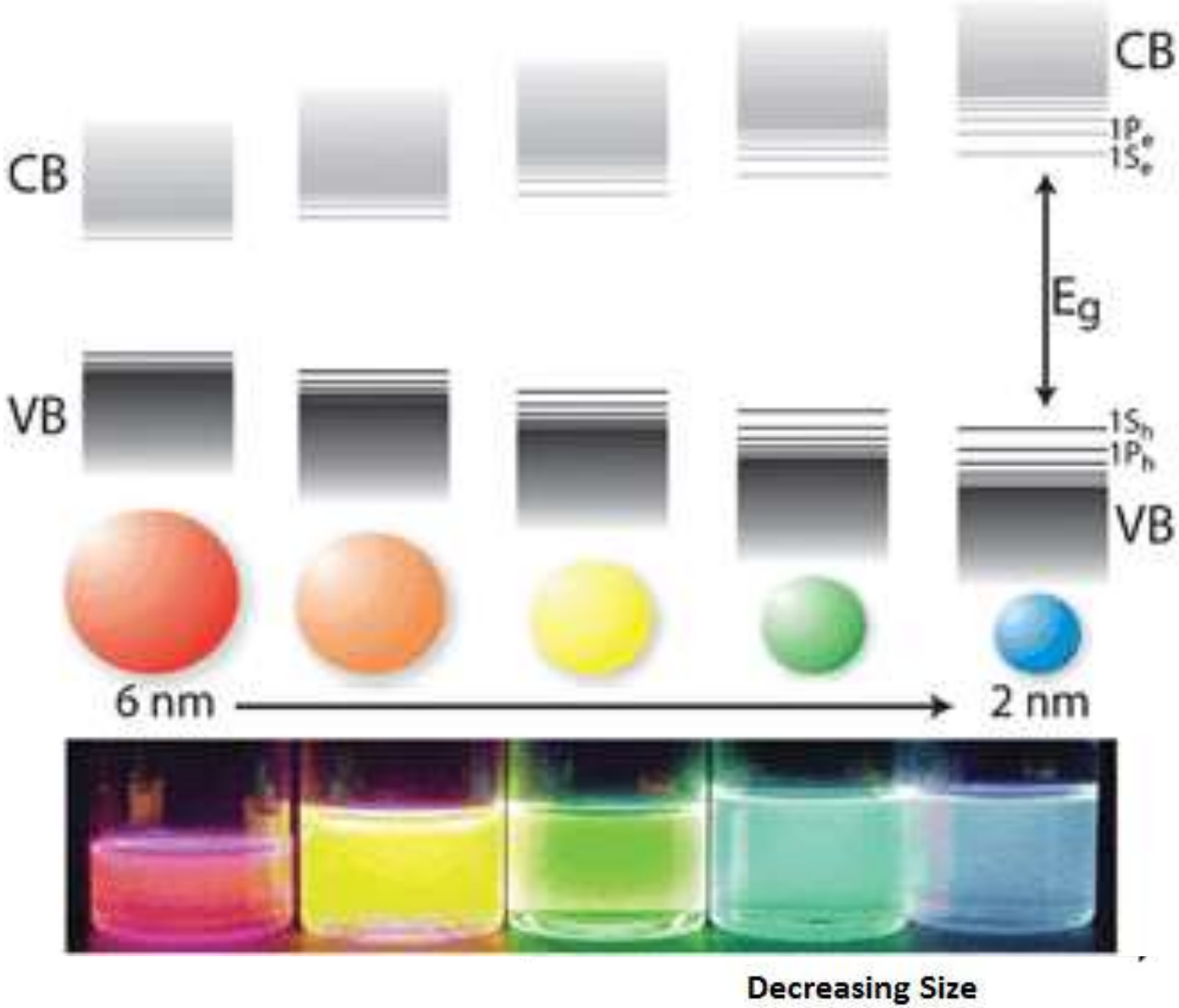
- When particles are created with dimensions of about 1–100 nanometers (where the particles can be “seen” only with powerful specialized microscopes), the materials’ properties change significantly from those at larger scales.
- Properties such as **Optical**, **electrical, magnetic, and chemical reactivity** change as a function of the size of the particle.
- Nanoscale materials have far **larger surface area**s than similar masses of larger-scale materials. As surface area per mass of a material increases, a ***greater amount of the material can come into contact with surrounding materials, thus affecting reactivity.***





Particle size

Band gap of nanomaterial increases with decrease in the size



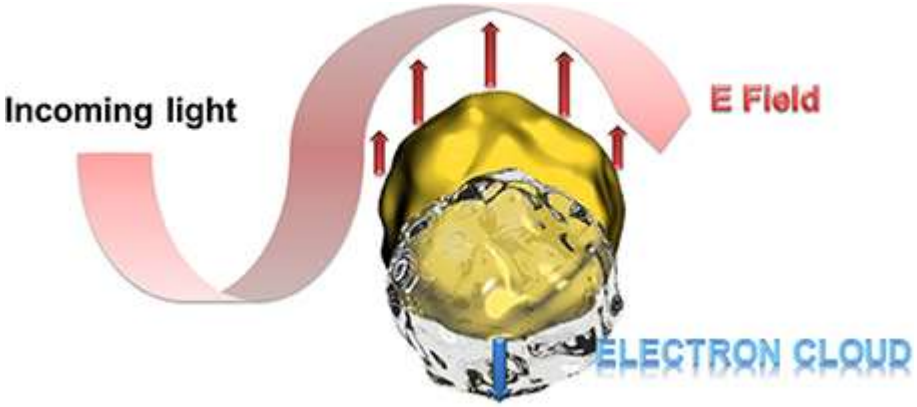
Optical Properties of Nanomaterials

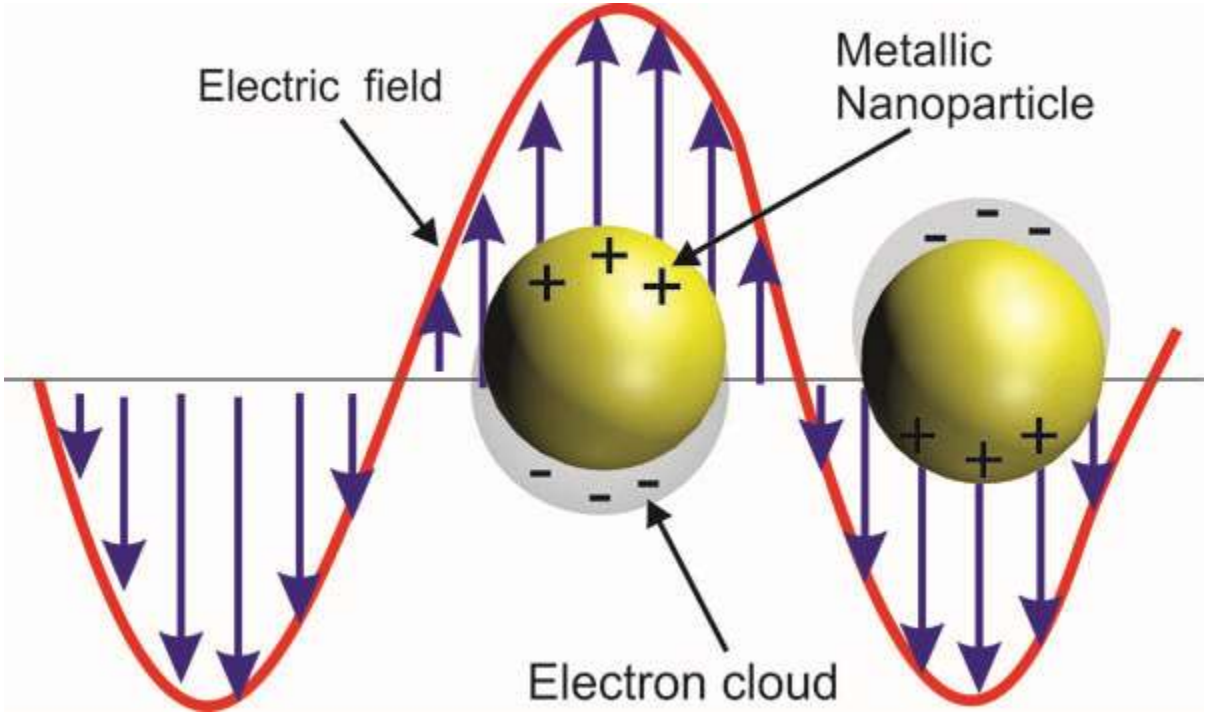
One of the most fascinating and useful aspects of nanomaterial is their ***optical properties.***

The change in optical properties is mainly due to the surface plasmon resonance and the increased energy level spacing.

Surface Plasmon Resonance (SPR): ***The coherent excitation of entire free electrons in the conduction band may produce an in-phase oscillation with incident visible light, called surface Plasmon resonance.*** *When the size of a metal nanocrystal is smaller than the wavelength of incident radiation, a surface plasmon resonance is generated. Thus, plasmon resonance depends* ***on the particle size****.*

In solid state physics*, the plasmon represents the* ***collective oscillation of a free charge in a metal****, and may be considered as a kind of* ***plasma wave****. The positive electrical charge in the metal* ***is fixed*** *and the free electron* ***is free to move*** *around it. An applied external electric field, as from a light source, causes the free electrons at the surface of the* ***metal to vibrate collectively****, giving rise to surface plasmons.*


Incoming light
E Field
ELECTRON CLOUD


Electric field
Metallic
Nanoparticle
Electron cloud

- *"Since electrons are also **particles with an electric charge**, when they vibrate they also **generate an electric field**, and when the electric field from the vibration of free electrons and the applied external electric field (e.g., electromagnetic waves) resonate the resulting phenomenon is referred to as a **surface plasmon resonance"** that takes place at the surface of the metal.*

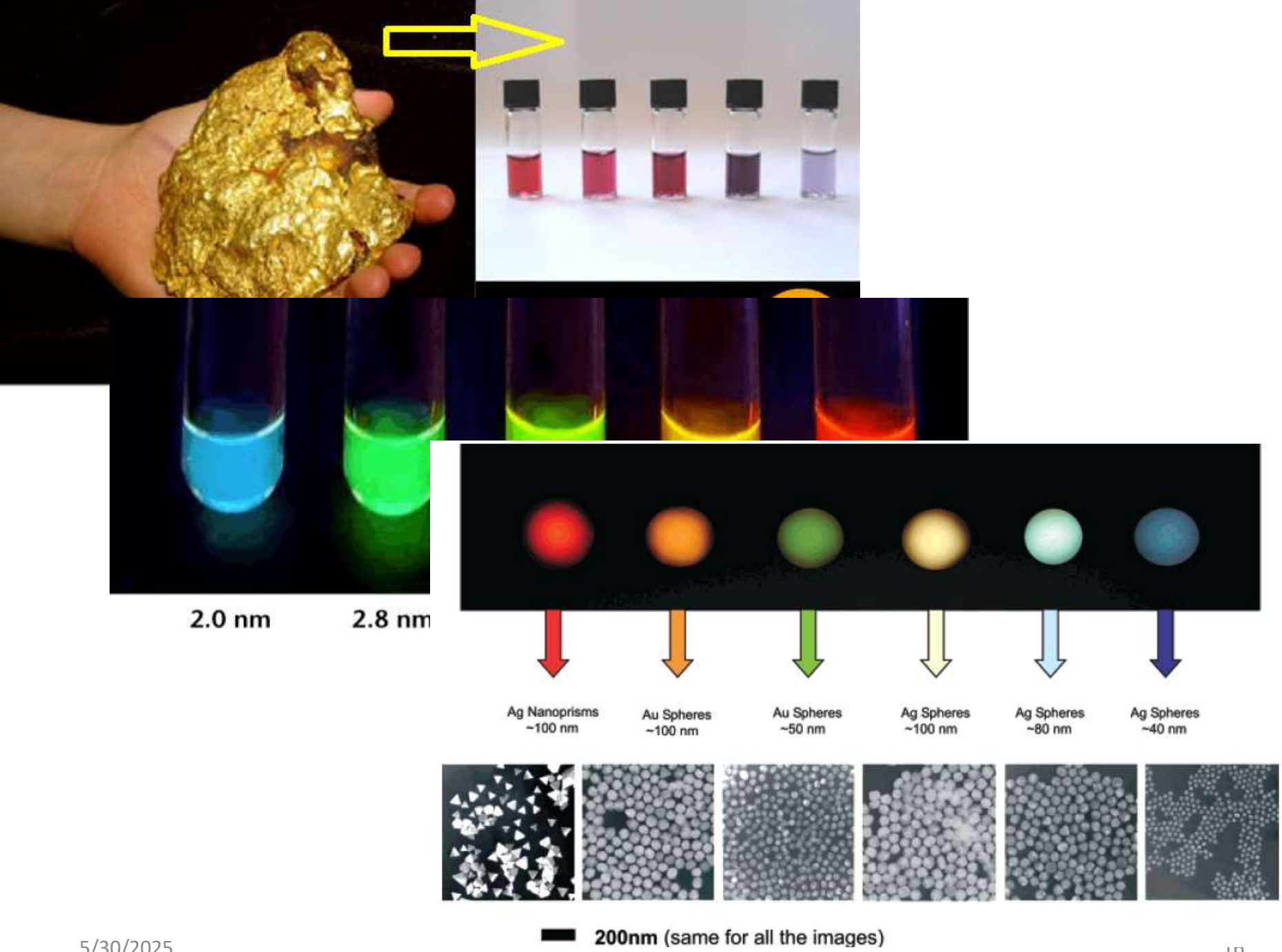
- *However, if light irradiates a solution that contains dispersed metal nanoparticles smaller than the wavelength of light, then depending on the electric field of light, the deviation produces a free electron at the surface of the metal. As a result, the weak or thick portions of the electric field appear on the nanoparticle surface and can be considered as a kind of polarization.*

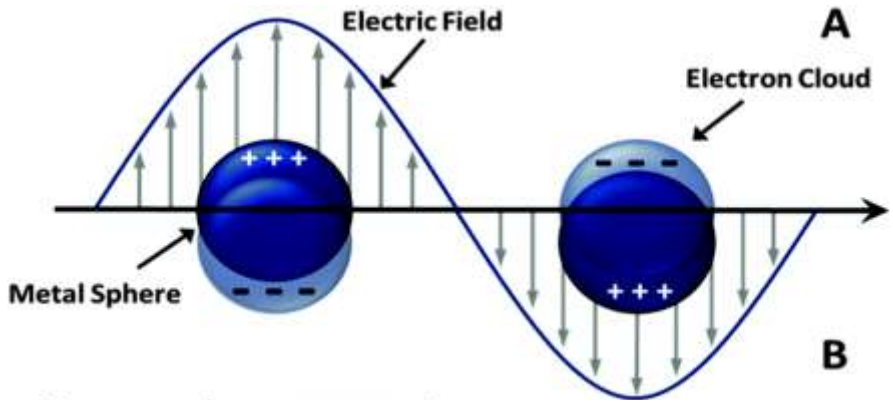
- Nanoparticles absorb visible light and produce surface plasmons.
- Small nanoparticles absorb blue-green wavelength of light, but they reflect red light.
- When size of the nanoparticles increases, wavelength of plasmon resonance absorption moves to longer wavelength (red wavelength) side. Now red light is absorbed and blue light is reflected, resulting in pale blue or purple colour for the particles.
- When particle size increase towards critical limit, SPR wavelength shifts to the IR spectrum of the radiation and visible lights are reflected. ***These properties find potential applications in biosensors***.

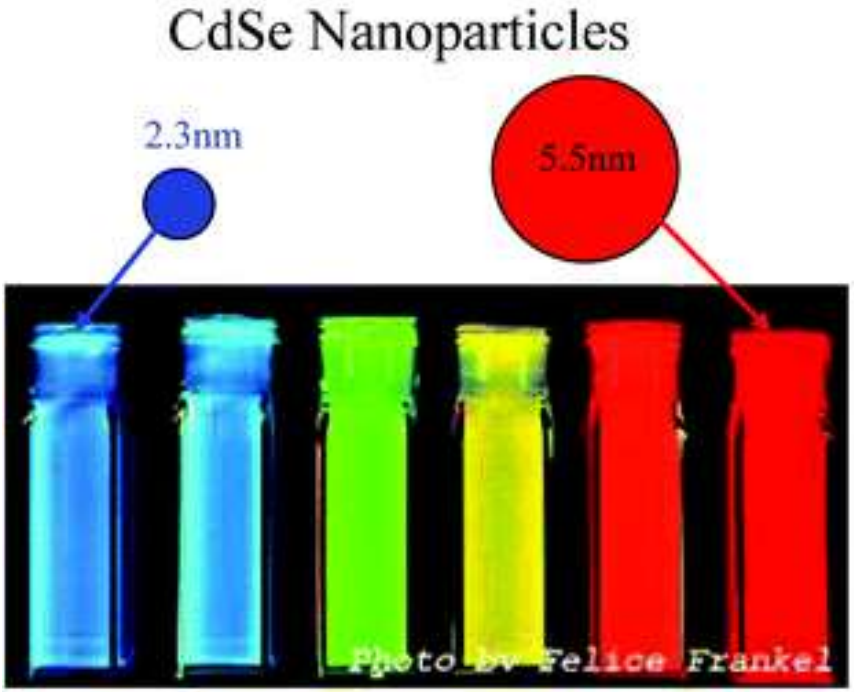
It should be noted that, metal nanorods exhibit two SPR modes, one due to **transverse** and the other due to **longitudinal** excitations. The wavelength of transverse mode is usually set at nearly **520 nm for gold and 410 nm for silver**. However, their longitudinal modes can be tuned to span from **visible region to near IR region** by controlling their aspect ratios.

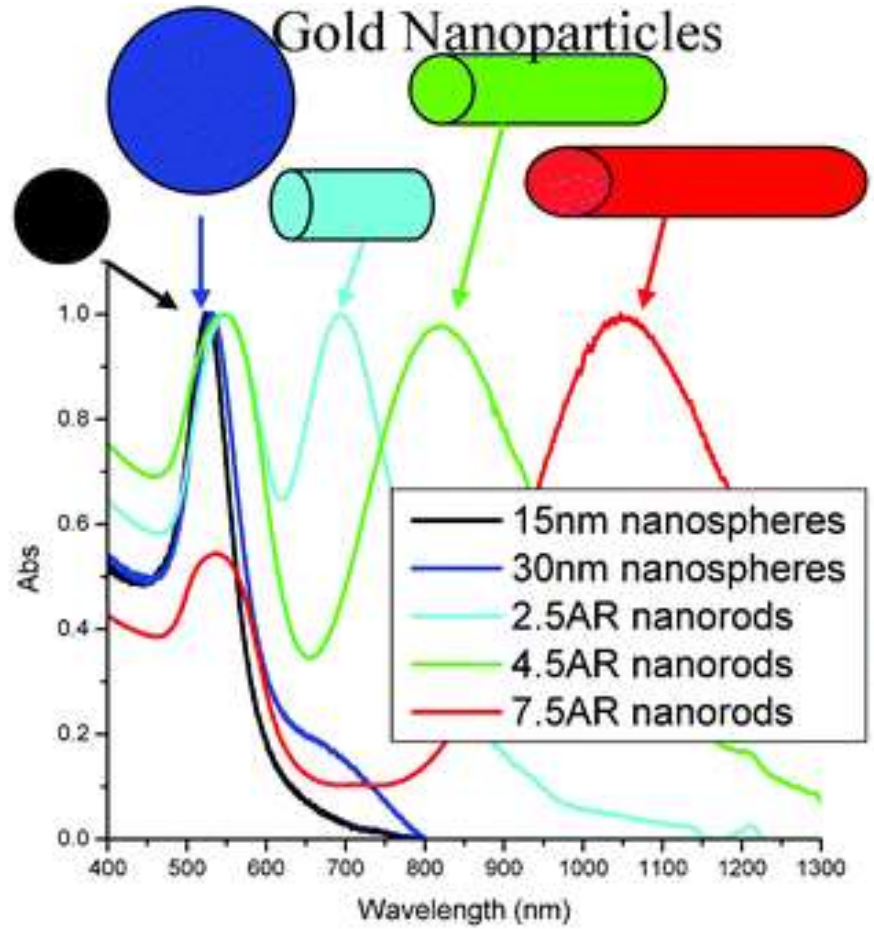
*The wavelength corresponding to the SPR depends on the **kind of metal**, the **shape of the metal nanoparticle,** and the **extent of aggregation of the metallic nanoparticles**.*

- *For example, the wavelength of the SPR band maximum of a spherical **Au nanoparticle is 520–550 nm**. If a colloidal Au nanoparticle solution is now irradiated with visible light at these wavelengths (520–550 nm), the visible light corresponding to the **green color** is absorbed and the particles now display a **red purple** color, which is the **complementary color to green**.*
- *In a colloidal **Ag nanoparticle** solution which has a SPR band maximum near **400 nm**, the **blue color** of the visible light is absorbed and the Ag nanoparticles now take on **a yellow color**, the complementary color to blue.*


2.0 nm
2.8 nm
Ag Nanoprisms ~100 nm
Au Spheres ~100 nm
Au Spheres ~50 nm
Ag Spheres ~100 nm
Ag Spheres ~80 nm
Ag Spheres ~40 nm
200nm (same for all the images)


Electric Field
A
Electron Cloud
Metal Sphere
B

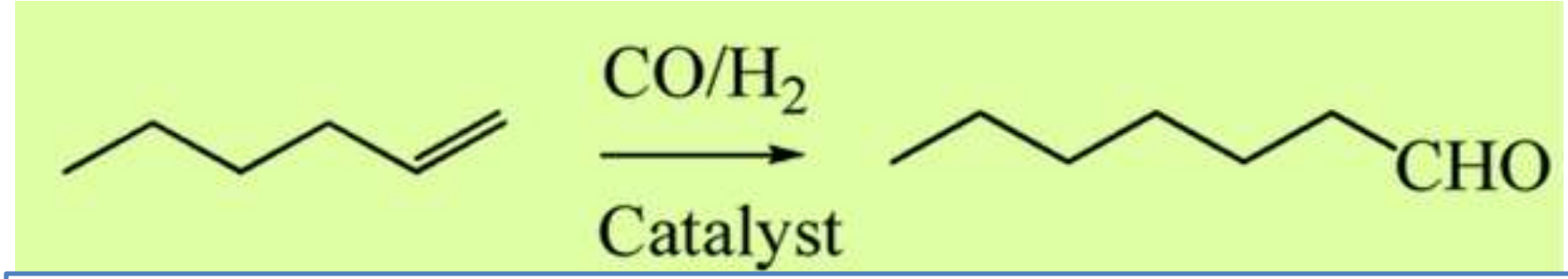

CdSe Nanoparticles
2.3nm
5.5nm
Photo by Felice Frankel


Gold Nanoparticles
Abs
Wavelength (nm)
15nm nanospheres
30nm nanospheres
2.5AR nanorods
4.5AR nanorods
7.5AR nanorods

Catalytic Properties of Nanomaterials

- The rate of any chemical reaction depends on the number of active reaction sites on the catalyst.
- The surface area of nanoscaled catalyst is larger than that of bulk material.
- As compared to bulk catalyst, the nanoscale catalyst of same material with same quantity has more number of active sites.
- Because of unsaturated valencies of surface atoms of Nanoparticle catalyst, they catalyze the reactions at much faster rates due to higher surface energies as compared to their respective bulk partners.
- Therefore, the nanoscaled material is found to exhibit very high catalytic efficiency than the corresponding bulk material.

Hydroformylation of 1–Hexene over Rh/Nano -oxide



- The highest activity with 100% total conversion and 96% yield of aldehydes was obtained with the **Rh/nano-ZnO catalyst.**
- The **Rh/nano-ZnO catalyst** was found to be more stable and active than the corresponding rhodium catalyst supported on **bulk ZnO (76% yield).**

Article

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Hydroformylation of 1-Hexene over Rh/Nano-Oxide Catalysts

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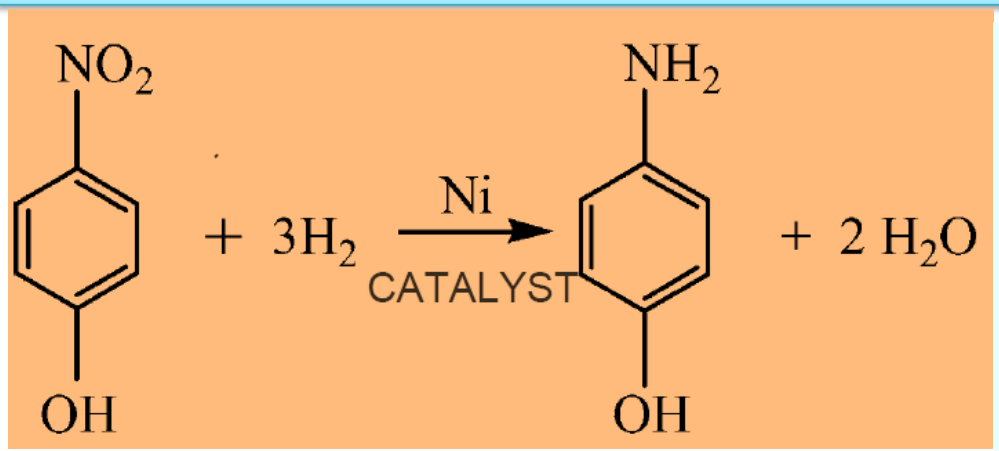
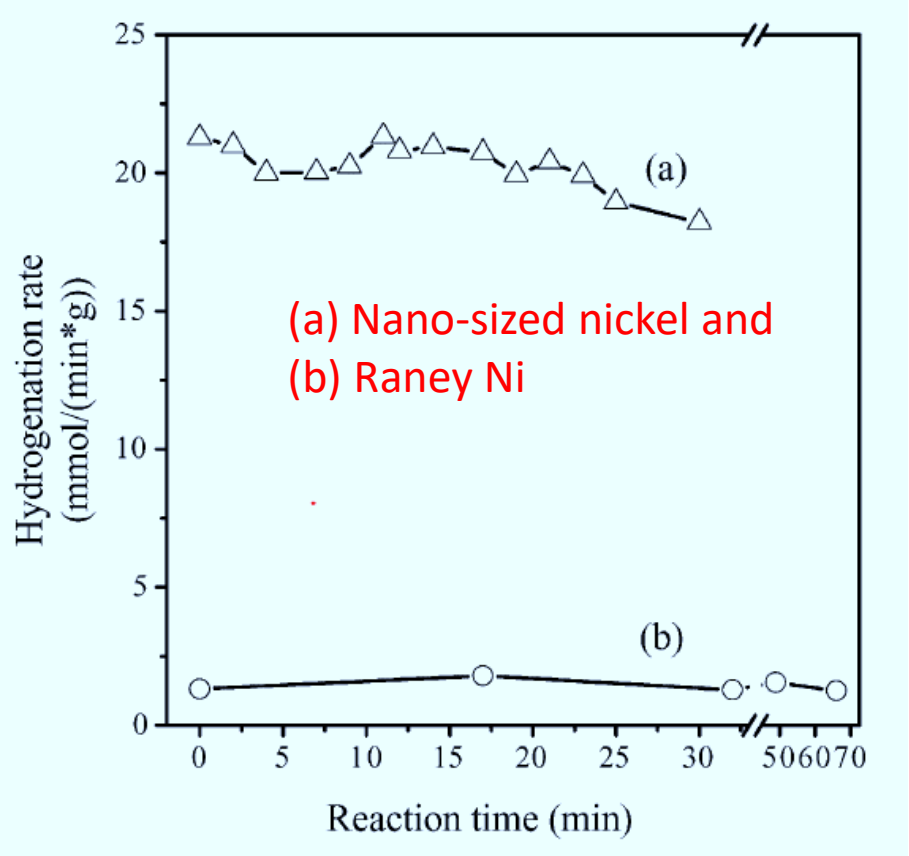
Maija-Liisa Kontkanen [1], Matti Tuikka [1,2], Niko M. Kinnunen [1], Sari Suvanto [1] and Matti Haukka [1,2,*]

Catalysts **2013**, *3*, 324-337; doi:10.3390/catal3010324

Catalytic hydrogenation of p-nitrophenol to p-aminophenol using nano-sized nickel





The hydrogenation rate of nano-sized nickel is about 15 times higher than that of Raney Ni at similar reaction conditions

The **nano-sized nickel** system suffers 54% deactivation, while the **Raney nickel** system suffers 80.4% deactivation only throughout six continuous hydrogenation cycles.

These results indicate that the catalytic stability of **nano-sized nickel** system is superior to that of the **Raney nickel system**

Synthesis of *p*-aminophenol from *p*-nitrophenol over nano-sized nickel catalysts

Yan Du, Hongling Chen*, Rizhi Chen, Nanping Xu

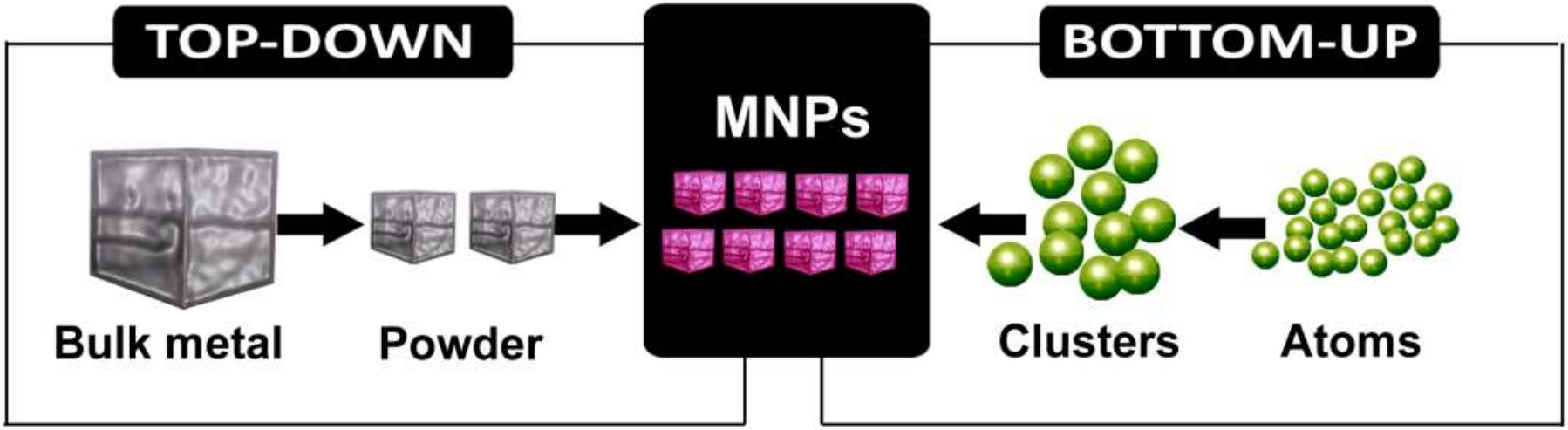
Applied Catalysis A: General 277 (2004) 259–264

Synthesis of Nanomaterials: SCS for metal oxide, Sol-Gel- for TiO_2 nanoparticles Carbon Nano materials

Researchers are active worldwide developing new preparation methods for functionally and technologically useful nanoparticles and nanostructures. Nature efficiently builds nanostructures by relying on chemical approaches. There are many methods developed to prepare nanomaterials, they are listed as chemical methods (bottom up) and physical methods (top down) as shown in Table.1

Table. 1 General methods of synthesis of nanomaterials

CHEMICAL METHODS (Bottom Up)	**PHYSICAL METHODS** (Top down)
Combustion Synthesis	Inert gas condensation
Hydrothermal	Sputtering
Sol-gel method	Molecular beam epitaxy
Micelles micro emulsion	Lithography
Single crystal growth	Ion beam technique
Colloidal methods	Chemical vapor deposition


TOP-DOWN
BOTTOM-UP
MNPs
Bulk metal
Powder
Clusters
Atoms

1) Low temperature solution combustion (SCS) method

This method was discovered by ***Prof. K. C. Patil*** when the mixture of $Al(NO_3)_3 \cdot 9H_2O$ and urea solution, rapidly heated around 500 ºC in a muffle furnace. The author observed that the solution mixture undergoes vaporization followed by vigorous ignition with an incandescent flame yielding voluminous white product which was identified as α-Al_2O_3.

Combustion method: is a low temperature, time saving, energy efficient, self propagating method, involving **spontaneous exothermic redox reaction** between **metal nitrate and organic fuel**, used to prepare nanomaterials.

Principle: The method is based on the utilization of heat energy produced during the exothermic spontaneous redox reaction between an oxidizer (metal nitrate) and a reducing agent (organic fuel). The oxidizer can be of any metal nitrates and reducing agents may be organic fuels, such as glycine, oxalic acid, urea, hexamine, sugar, EDTA, Dextrose etc.

Metal nitrate + Fuel $\longrightarrow$ **nano Metal oxide + Gases**

The amount of fuel can be calculated in such way that the total oxidizing valency of fuel should match with the total reducing valency of metal nitrate. The following formula can be used to calculate the amount of fuel:

$$\frac{\text{Wt. of fuel}}{\text{Mol. Wt. of fuel}} \times \text{Valency of fuel} = \frac{\text{Wt. of metal nitrate}}{\text{Mol. Wt. of metal nitrate}} \times \text{Valency of metal nitrate}$$

For the calculation of valency of fuel and metal nitrate, the valency to be considered for the elements such as of N, O, H, C, and M^{n+} are 0, -2, +1, +4 and +n respectively.

Procedure:

Step1: Weighing and mixing of the reactants

Weighing of all the reactants (metal nitrate and a fuel) according to stoichiometry, by using the above formula and dissolved in minimum quantity of distilled water in a beaker of suitable size.

The mixture is stirred on a magnetic stirrer till to get uniform solution as shown in figure (left side).

Magnetic stirrer on which uniform redox solution is obtained by stirring

Step 2: Evaporation and combustion

The uniformly mixed solution is kept in a furnace maintained at 500 °C

Evaporation (boiling) of water

Spontaneous/vigorous combustion occurs and propagates throughout the redox mixture

Final nano powder of corresponding metal oxide

❖The clear solution is kept in a furnace (shown below) maintained at 500 ºC.

❖Firstly the solution undergoes evaporation that results in more concentrated, uniformly mixed viscous-gel type substance.

❖After some time, the viscous-gel catches fire and propagates spontaneously in the redox mixture in the form of either a flame type or smouldering type.

❖The combustion lasts for about 1-2 min.

❖During the flame propagation large quantity of gasses and high temperature produced helps in the formation of respective nano metal oxide.

❖The SCS has emerged as a viable technique for the preparation of advanced nano-metal oxide.

The **most important advantages of SCS process are:**

✓It is an easy and fast process that uses relatively simple equipment.

✓High-purity products can be easily prepared by using this method.

✓Composition, structure, homogeneity, size and stoichiometry of the products can be controlled.

✓High exothermicity of the metal nitrate–fuel reaction permits incorporation of desired quantity of dopants in the various hosts of industrially useful phosphors.

✓This is the cheap method as compared to conventional solid state method

Stoichiometry calculation

$$\frac{\text{Wt. of fuel}}{\text{Mol. Wt. of fuel}} \times \text{Valency of fuel} = \frac{\text{Wt. of metal nitrate}}{\text{Mol. Wt. of metal nitrate}} \times \text{Valency of metal nitrate}$$

Valency calculation					
$Zn(NO_3)_2 \bullet 6H_2O$ (Zinc Nitrate) 297 g/mol	Zn2+	2N	6O	6H2O	**Total Valency**
	+2	2x0=0	6x(-2)=-12	6x0=0	**-10**
$Al(NO_3)_3 \cdot 9H_2O$ (Aluminium nitrate) **375.14** g/mol	A3+	3N	9O	9H2O	
	+3	3x0=0	9x(-2)=-18	9x0=0	**-15**
C6H12O6 (Glucose,) 180.156 g/mol	6C	12H	6O		
	6x(+4)=+24	12x(+1)=+12	6(-2)=-12		**+24**
NH_2-CH_2-COOH (Glycine) 75.07 g/mol	N	2O	5H	2C	
	1x0=0	2x(-2)=-4	5x(+1)=+5	2x(+4)=+8	**+9**

Calculate the amount of glycine fuel required to

prepare nano ZnO using

8g of hydrated Zinc nitrate (Mol. Mass =297g/mol)

[(Wt x Valency)/Mol Mass] of Glycine = [(Wt x Valency)/Mol Mass] of Zinc nitrate

[(Wt x 9)/75.07] of glycine = [(8x10)/297]

Wt of glycine =…….. g

2. Sol-gel method

It is a wet chemical technique widely used for the fabrication of nano-structured ceramic materials and thin films.

Sol-gel process involves the conversion of precursor solution (usually metal salts or metal alkoxide) into a nano-structured inorganic solid through inorganic polymerization reactions catalyzed by water.

In general, metal alkoxides (M-OR) are widely used as precursors, because they readily react with water.

The reactions involved in the sol-gel process are

1) *Hydrolysis*

$$\underset{\text{Metal alkoxide}}{M\text{-}OR} + H_2O \xrightarrow{\text{Ethanol}} \underset{\text{Metal hydroxide}}{M\text{-}OH} + ROH$$

2) Condensation

$$M\text{-}OR + H\ O\text{-}M \longrightarrow M\text{-}O\text{-}M + R\text{-}OH$$

Condensation…… Condensation ……. Condensation ……of …..M-O-M results in to a polymer network in all the possible directions.

Process:

Step 1: Preparation of homogeneous solution either by dissolution of metal salt in water or metal alkoxide in an organic solvent (usually alcohol) that is miscible in water.

Step 2: 'Sol' formation by hydrolysis: It involves the conversion of homogeneous solution into a "sol". (A stable dispersion of colloidal particles of precursors in a solvent is known as sol) due to hydrolysis reaction. During hydrolysis, the alkoxide groups (-OR) are replaced via the nucleophilic attack of the oxygen atom of a water molecules, results in release of alcohol and formation of metal hydroxide (sol).

Step 3: 'Gel' formation by condensation

The colloidal solution is kept for aging. During aging condensation reaction between two metal hydroxyl/alkoxy species leads to M-O-M bonds with the release of H_2O/R-OH. This condensation process continues and finally results in a "gel", an interconnected, a rigid and porous inorganic network covered completely with liquid phase. This transformation is called **Sol-Gel transition.**

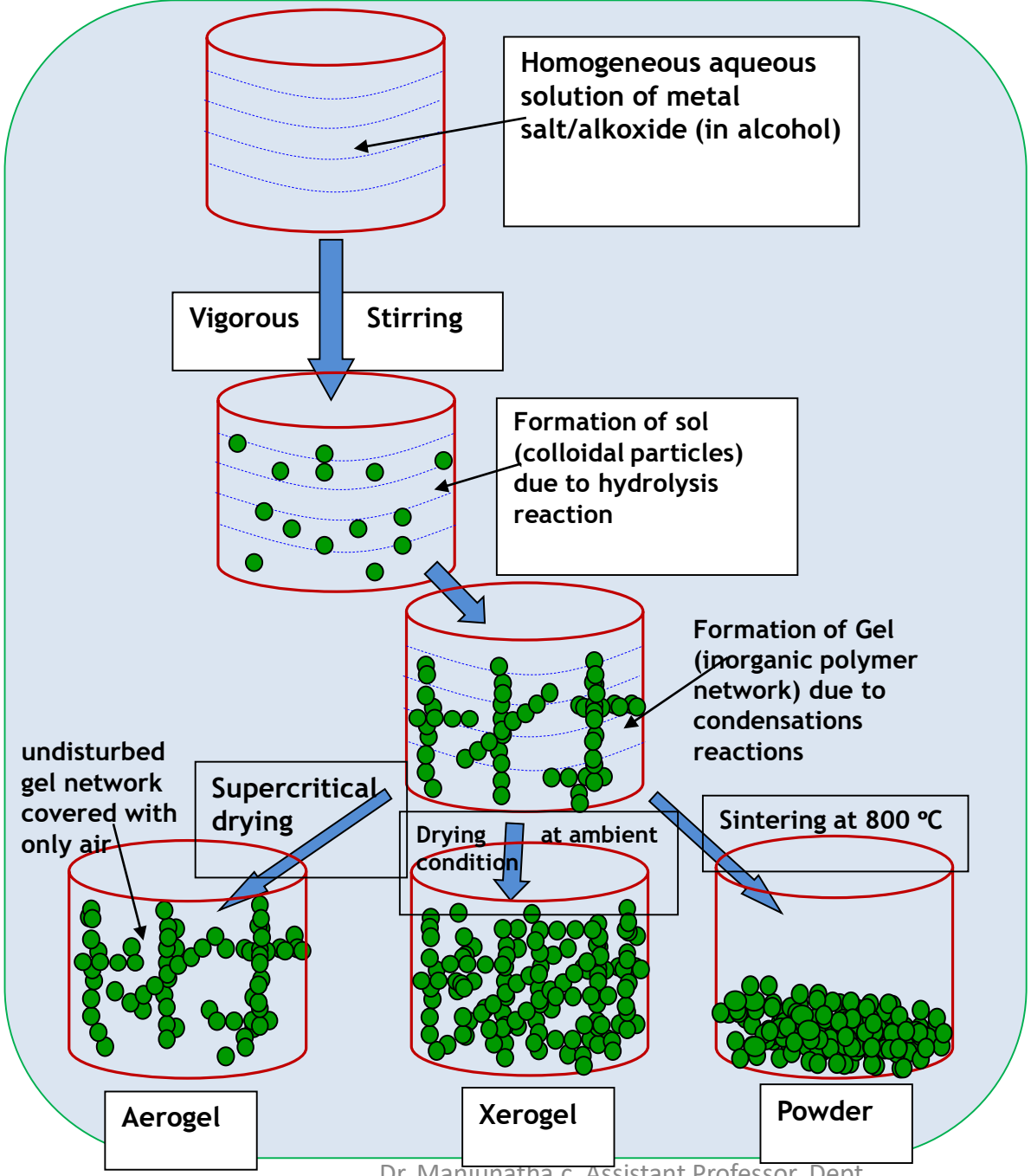
Step 4: Drying of gel:

It involves removal of liquid phase from gel network. There are different ways of drying gel:

•If the gel medium is dried, by removing the liquid solvent (under hyper critical conditions) without destroying the gel network, **"aerogel" is** produced.

• If the solvent is dehydrated by under ambient conditions (removal of R-OH groups), **"xerogel"** are produced.

•If the gel network is sintered at high temperature (800ºC), densification, decomposition of gel results in complete collapse of gel net into powder.


Homogeneous aqueous solution of metal salt/alkoxide (in alcohol)
Vigorous
Stirring
Formation of sol (colloidal particles) due to hydrolysis reaction
Formation of Gel (inorganic polymer network) due to condensations reactions
undisturbed gel network covered with only air
Supercritical drying
Drying at ambient condition
Sintering at 800 °C
Aerogel
Xerogel
Powder

Example: Preparation of TiO_2 by Sol-gel method:

Chemicals required: Titanium isopropoxide [$Ti(O(CH(CH_3)_2)_4$], absolute alcohol, distilled water, HNO_3

Equipments/glassware:

• 250 ml glass beaker, magnetic stirrer, Furnace, oven.

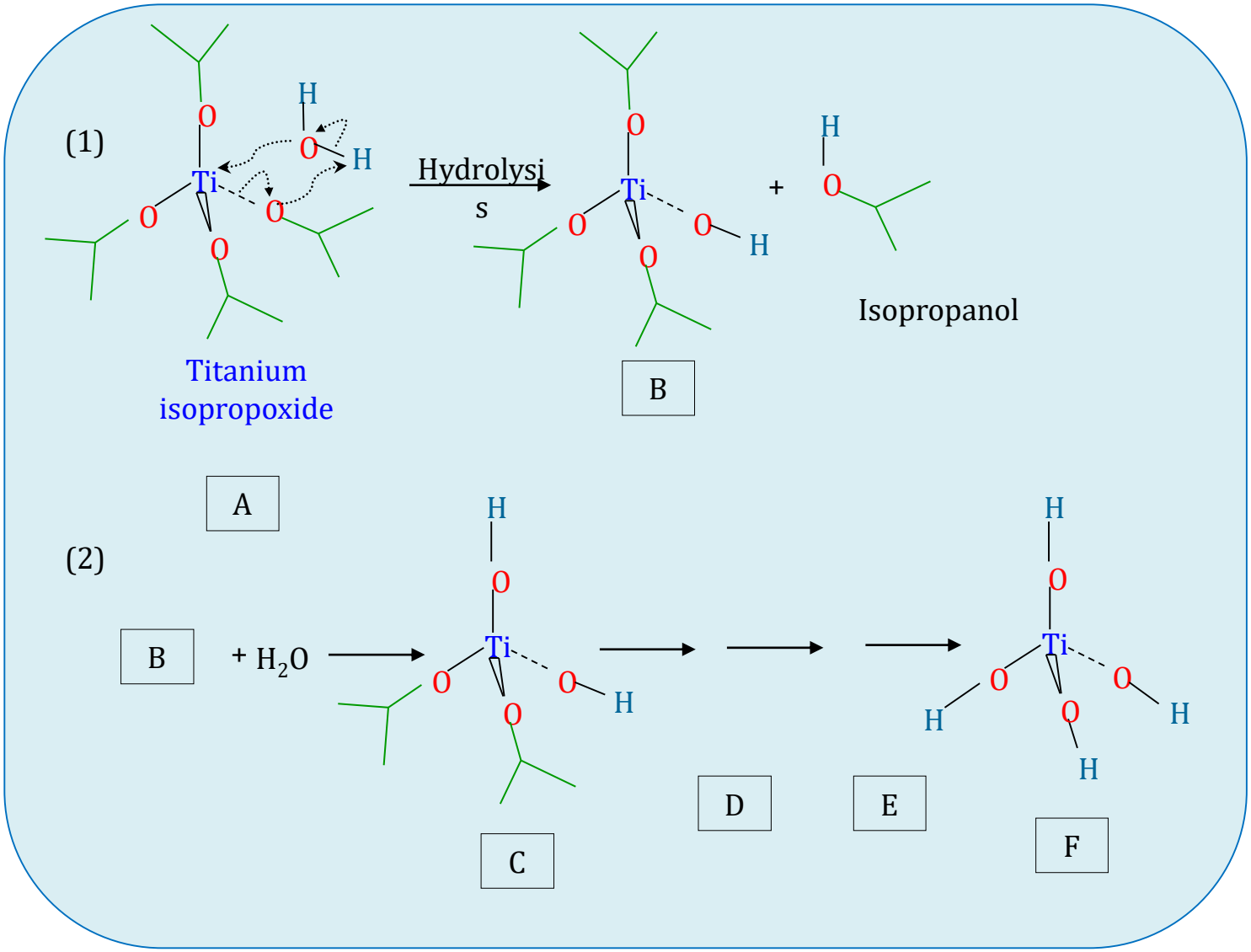
Procedure: Stoichiometric quantity of titanium (IV) isopropoxide is dissolved in absolute ethanol and distilled water in the ratio of $Ti:H_2O=1:4$. HNO_3 is added to adjust pH and restrain the hydrolysis process of the solution.

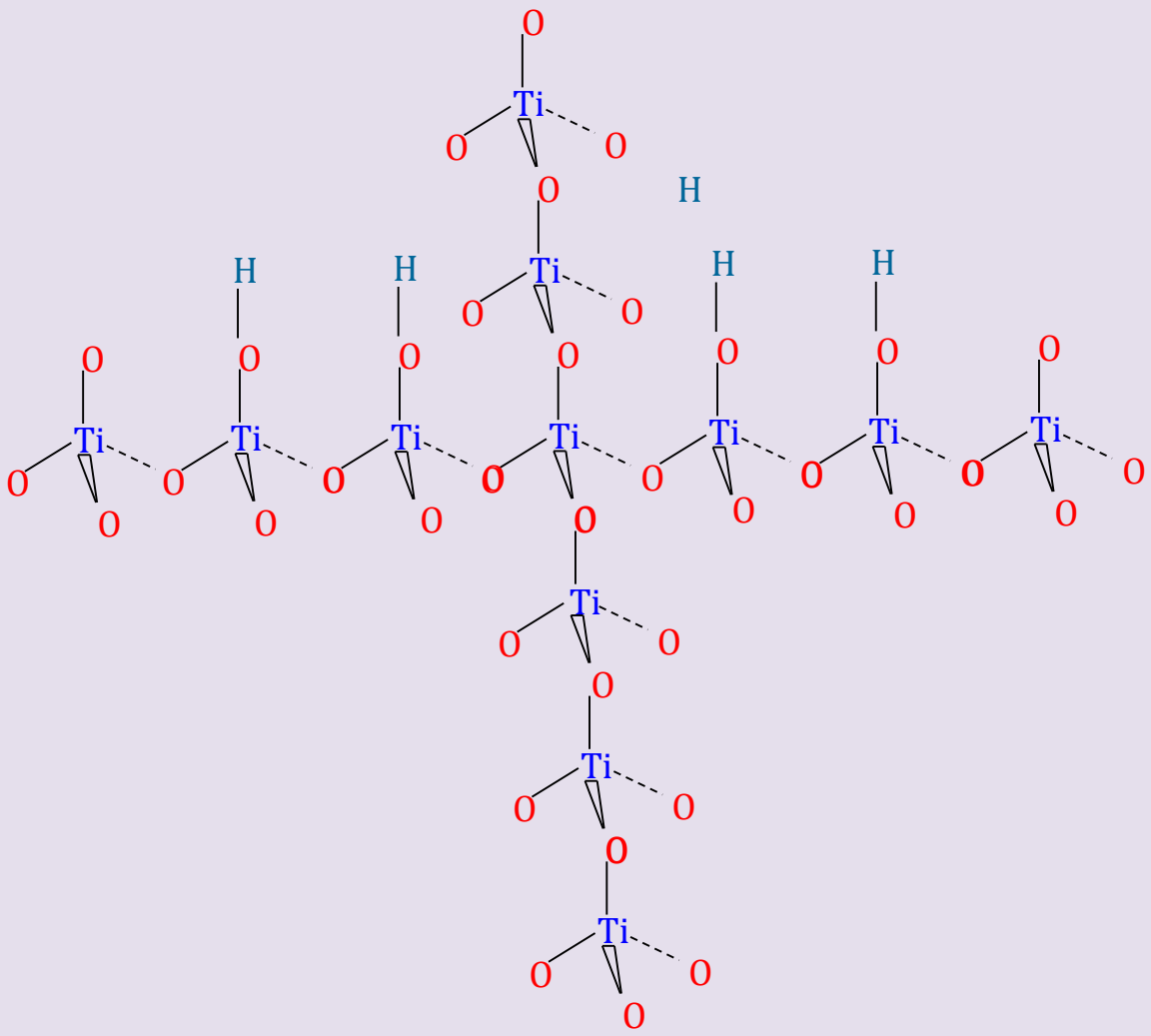
•The solution is vigorously stirred for 30 min in order to form sol (a suspension of colloidal particle).

•The sol is kept for aging for about 24 h. During aging sol is transformed into gel. (It is due to polycondensation and formation of inorganic network).

•In order to obtain TiO_2 nanoparticles, the gels are dried at 120 ºC for 2 h to evaporate water and organic solvent. Then the dry gel is sintered at 450 ºC for 2 h to obtain TiO_2 nanoparticles.

***Reaction mechanism**: Nucleophilic (H_2O) attack on alkoxy group.*





The advantages of this technique are evident.

- ❖It is a simple and economic technique, as the fabrication does not need expansive machinery or apparatuses.
- ❖It offers a high flexibility as one can produce materials with a wide range of stoichiometry and additional dopants.
- ❖It allows the fabrication of high quality coatings.
- ❖The starting materials are easily to obtain, not expensive and available in a high purity.
- ❖Nanomaterials of variety of shapes can be prepared.